

Name: solutions (please print)

Signature: _____

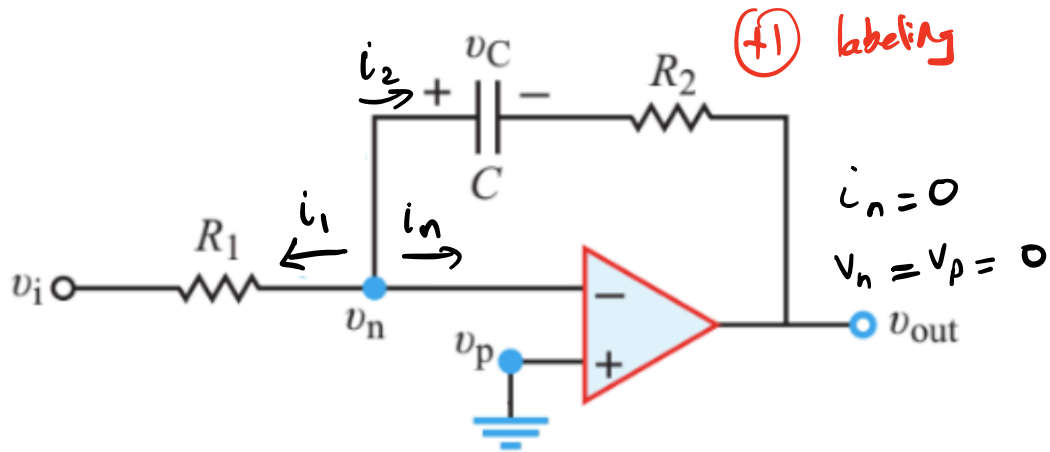
ECE 2202 – Quiz 3

October 7, 2025

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
3. You will have 40 minutes to work on the quiz.
4. Write your ID number on every page that you show your work.

_____/20

Problem 1 (8 pts)

Relate v_{out} to v_i in the circuit. Assume $v_c = 0$ at $t = 0$.

(+2) ① KCL @ v_n : $i_1 + i_2 + i_n = 0 \Rightarrow i_1 = -i_2$

② $i_1 = \frac{v_n - v_i}{R_1}$

③ $v_n - v_{out} = v_c + i_2 \cdot R_2$

④ $i = C \frac{dv}{dt}$

⑤ $-v_{out} = \frac{1}{C} \int_0^t i_2 dt + i_2 \cdot R_2$

$v_c(t) = \frac{1}{C} \int_0^t i_2 dt + v_c(0)$ (+2)

⑥ $i_2 = -i_1 = -\left(\frac{-v_i}{R_1}\right) = \frac{v_i}{R_1}$

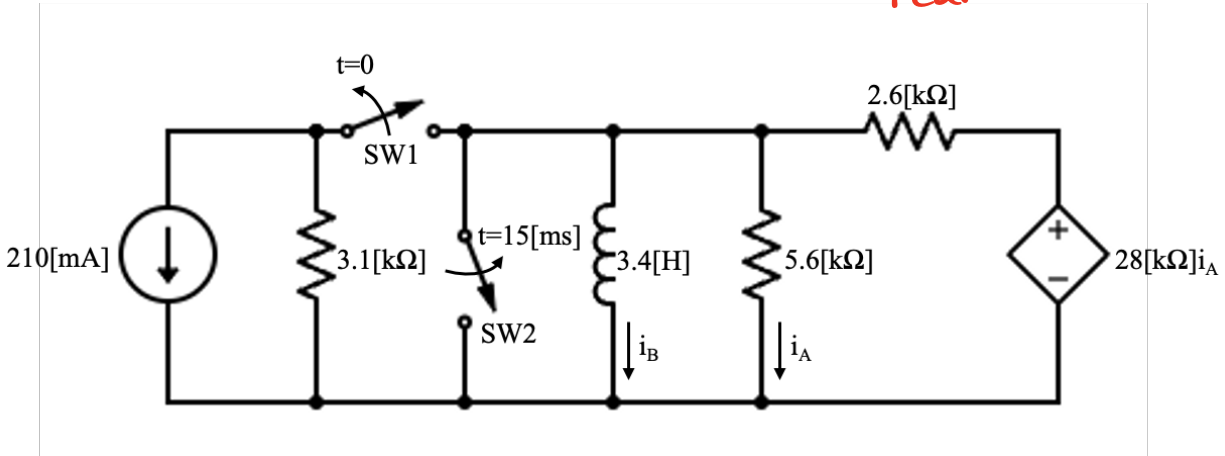
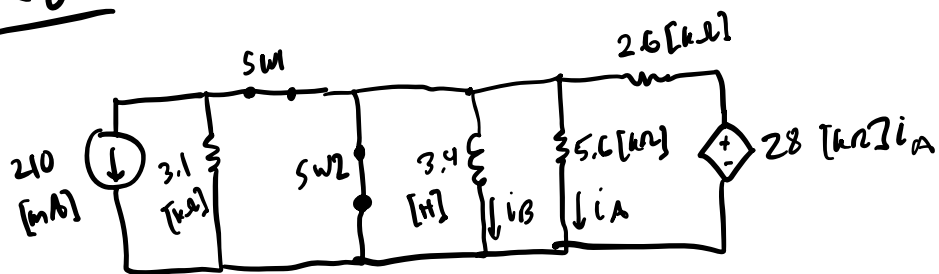
⑦ $v_{out}(t) = -\left(\frac{1}{R_1 C} \int_0^t v_i dt + i_2 R_2\right)$ (+3)

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Problem 2 (12 pts)

The two switches, SW1 and SW2, in the circuit below were both closed and the current $i_B(t)$ was 33[mA], for $t < 0$. Then, at $t = 0$, switch SW1 opened. Following that, at $t = 15$ [ms], switch SW2 opened. Find $i_A(20$ [ms]).

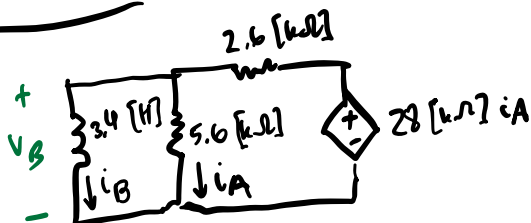
redraw circuits (+3)

 $t < 0$:

for all $t < 15$ [ms], zero voltage across inductor (parallel w/ short),
so i_B is constant through the whole time:

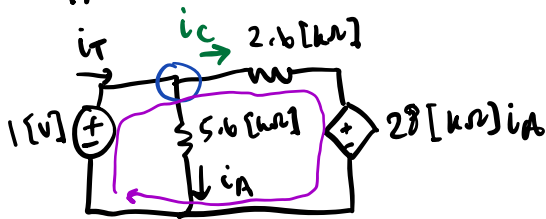
$$i_B(t) = 33 \text{ [mA]} \text{ for } t < 0 \quad (+1)$$

$$i_B(t) = 33 \text{ [mA]} \text{ for } t \leq 15 \text{ [ms]}$$

 $t > 15$ [ms]:

this looks like a natural response circuit $\rightarrow$ we already have the initial condition for i_B , now we need the time constant, i.e. the equivalent resistance seen by the inductor, R_{eq}

apply a test source:



$$i_A = \frac{1 \text{ [V]}}{5.6 \text{ [k}\Omega]} = 0.179 \text{ [mA]}$$

(KCL) $-i_T + i_A + i_C = 0$

(KVL) $-1 \text{ [V]} + i_C \cdot 2.6 \text{ [k}\Omega] + 28 \text{ [k}\Omega] i_A = 0$

$$i_C = \frac{1 - 28 \text{ [k}\Omega] i_A}{2.6 \text{ [k}\Omega]}$$

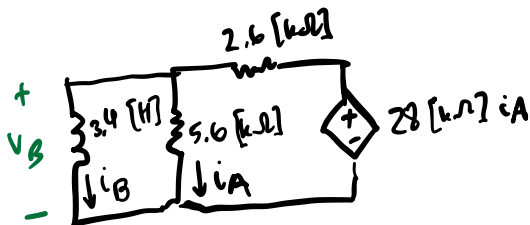
$$-i_T + i_A + \frac{1 - 28 \text{ [k}\Omega] i_A}{2.6 \text{ [k}\Omega]} = 0 \Rightarrow i_T = -1.36 \text{ [mA]}$$

$$R_{eq} = \frac{1 \text{ [V]}}{i_T} = -735 \text{ [}\Omega] \Rightarrow \tau = \frac{3.4 \text{ [H]}}{R_{eq}} = -4.62 \text{ [ms]}$$

solve for
natural
response
+4

$$i_B(t) = 33 \text{ [mA]} e^{-(t-15 \text{ [ms]}) / -4.62 \text{ [ms]}} \text{ for } t > 15 \text{ [ms]}$$

To figure out $i_A(20 \text{ [ms]})$ we can define v_B across the inductor



$$v_B(t) = 3.4 \text{ [H]} \cdot \frac{di_B(t)}{dt} \quad \left(L \frac{di(t)}{dt} \right)$$

$$v_B(t) = 24.27 \text{ [V]} e^{(t-15 \text{ [ms]}) / 4.62 \text{ [ms]}} \text{ for } t > 15 \text{ [ms]}$$

then, $i_A(20 \text{ [ms]}) = \frac{v_B(20 \text{ [ms]})}{5.6 \text{ [k}\Omega]}$

$$v_B(20 \text{ [ms]}) = (24.27 \text{ [V]}) e^{(20-15) / 4.62} = 71.6 \text{ [V]}$$

$$i_A(20 \text{ [ms]}) = \frac{71.6 \text{ [V]}}{5.6 \text{ [k}\Omega]} = \boxed{12.8 \text{ [mA]}} \quad (+2)$$

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$$i = \frac{dq}{dt} \quad v = \frac{dw}{dq} \quad p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = iv$$

$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$v = iR$$

$$R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

$$p = iv = i^2 R = \frac{v^2}{R}$$

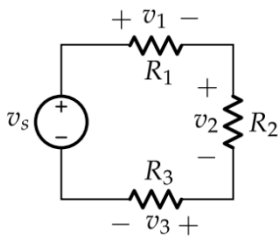
$$w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

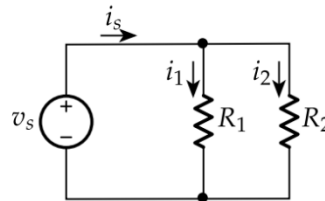
(Switch action at $t = T_0$)



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3} v_s$$

$$v_2 = \frac{R_2}{R_1 + R_2 + R_3} v_s$$

$$v_3 = \frac{R_3}{R_1 + R_2 + R_3} v_s$$



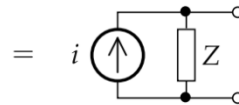
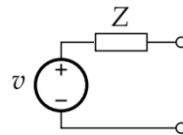
$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

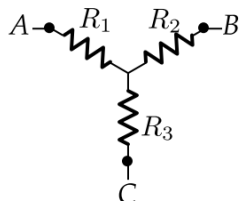
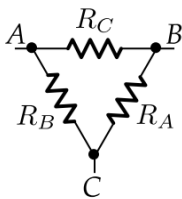
$\bar{V}_b \rightarrow$ Phasor notation

$$\text{CMRR} = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$



$$v = i \cdot Z$$

$$i = \frac{v}{Z}$$



$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$